

THAMARAISELVAN M

Software Engineer | Python | Node.js | AI/ML

✉ santhoshtamil1918@gmail.com ☎ +91 6369600176 📍 Coimbatore, Tamil Nadu

🌐 thamarai-portfolio-zc56.vercel.app/ 💼 linkedin.com/in/thamarai-selvan-ts-baa3601a1

PROFESSIONAL SUMMARY

Software Engineer with 4+ years of software development experience, focused on Python and Node.js. Experienced in building Python applications using FastAPI, Flask, and Django, including REST APIs, backend services, and database integrations. Hands-on experience using Python for data preprocessing, predictive modeling, and computer vision with Pandas, NumPy, Scikit-learn, and OpenCV. Skilled in developing Node.js and Express.js services with authentication, payment integrations, and real-time communication using SocketIO. Experienced in SQL databases, application debugging, deployment, and production support.

PROFESSIONAL EXPERIENCE

Backend Developer

02/2022 – 03/2023

Squash Apps Pvt. Ltd.

Coimbatore

- Developed Node.js application functionality, including API endpoints and business logic.
- Implemented database operations to support application data management.
- Collaborated with frontend developers to integrate APIs and resolve application issues.

Software Engineer

05/2023

KnowledgeQ Interactive Consultancy Services Pvt. Ltd.

Coimbatore

- Developed application services using Node.js and Python, implementing APIs, business logic, and database integrations.
- Built machine learning and computer vision components for forecasting and manufacturing inspection workflows.
- Integrated prediction and image-processing outputs with applications to support automated reporting.
- Supported application integration, debugging, and deployment.

TECHNICAL SKILLS

Languages

JavaScript, Python, SQL

Frameworks

Node.js, Express.js, FastAPI, Flask, Django

Software Development

REST APIs, Microservices, Object-Oriented Programming, Database Design

Databases

MySQL, SQLite, PostgreSQL

Authentication & Integration

JWT, RBAC, OAuth 2.0, Webhooks, Razorpay, Easebuzz

Real-Time Systems

Socket.IO, WebSockets

AI & Machine Learning

Scikit-learn, TensorFlow, Keras, OpenCV, Pandas, NumPy

Cloud & Tools

AWS EC2, S3, CloudWatch, Docker, Nginx, Git, GitHub, Linux

Testing

Postman, Swagger/OpenAPI, PyTest, API Testing, Debugging

Frontend Fundamentals

HTML, CSS, JavaScript, Bootstrap

SELECTED PROJECTS

ERP Application (Squash Apps)

Node.js, REST APIs, MySQL

- Developed services for HRM, CRM, warehouse, and inventory management.
- Implemented employee, customer, order, and inventory workflows; optimized MySQL queries and integrated APIs with the frontend.

FinTech Payment Platform (KnowledgeQ)

Node.js, REST APIs, Razorpay, Easebuzz

- Independently designed, developed, and maintained the backend architecture for payments, user management, transactions, and notifications.
- Supported **₹11 crore+ in real transactions**, with the platform continuing to process payments in production.
- Integrated payment initiation, verification, refunds, and webhook processing.
- Implemented authentication, authorization, and logging; managed production deployments, monitoring, and troubleshooting.

Ship Management System (KnowledgeQ)

Socket.IO, WebSockets, REST APIs

- Built APIs to manage vessel, crew, voyage, and operational information.
- Implemented real-time operational updates and optimized concurrent event handling.

Ships Profit and Loss Management System (KnowledgeQ)

Node.js, Socket.IO

- Independently developed and maintained the complete backend for managing ship profit and loss information.
- Implemented real-time data updates using Socket.IO .
- Managed the application database and implemented data storage, retrieval, and updates to support financial workflows.

LPG Booking Prediction (KnowledgeQ)

Python, Scikit-learn

- Developed models to predict customers' next cylinder booking dates using historical booking data.
- Performed data preprocessing, feature engineering, and model evaluation; integrated predictions into application APIs.

Trucks Cabin Inspection System (KnowledgeQ)

Python, OpenCV, CNN, Machine Learning

- Developed computer vision solutions for cabin quality inspection and manufacturing defect detection.
- Prepared datasets, trained and evaluated models, and integrated inspection pipelines with application reporting.

Manufacturing Machine Service Date Prediction (KnowledgeQ)

Python, Deep learning, XGBoost, Random Forest, Dash

- Developed a machine learning application to predict the next service date for manufacturing machines and support maintenance planning.
- Used XGBoost and Random Forest to model service-date predictions.
- Built a Dash dashboard to display predictions and help users review upcoming maintenance needs.